Skanda Shreesha Prasad

M.S. Computer Science Student

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Education

M.S. **New York University - Courant Institute of Mathematical Sciences**, Computer Science — New York, NY, July 2025 – present

- Coursework: Deep Learning, Computer Vision, Bayesian Machine Learning, Fundamental Algorithms, Operating Systems

B.Tech.PES University, Computer Science and Engineering — Bengaluru, India, Sept 2021 – May 2025

Research Experience

- **Hierarchical Planning, H-JEPA and World Models**, New York University, Prof. Yann LeCun and Prof. Jean Ponce, Mar 2026 - Present. Researching hierarchical JEPA world models for planning and reasoning, developing dynamic temporal abstraction methods, and evaluating hierarchical latent planning on robotic manipulation environments.
- **Multiscale JEPA for Self-Supervised Active Matter Representation Learning**, New York University, Prof. Yann LeCun and Prof. Alfredo Canziani, Apr 2026. Built global, spatial, and multiscale JEPA architectures under a 100M-parameter constraint; placed second on the course leaderboard and presented the project as an Outstanding Project.
- **Multimodal Protein Retrieval from Sequences and Cryo-EM Density Maps**, New York University, Prof. David Fouhey, Feb 2026 - Present. Developing joint embeddings for amino-acid sequences and 3D cryo-EM density maps using a JEPA-style cross-modal predictive objective.
- **Active Spectral Learning for Non-Stationary Gaussian Processes**, New York University, Prof. Andrew Gordon Wilson, Sep 2025 - Dec 2025. Built an active-learning pipeline with Generalized Spectral Mixture kernels and compared epistemic-variance acquisition with BALD and other baselines.
- **Code-Mixed LLM for Natural Language Generation and Translation**, PES University, Prof. Dr. Uma D, Oct 2023 - Jan 2025. Built an English-to-Hindi-English code-mixed translation system and developed English-Kannada code-mixing data for low-resource language modeling.

Experience

The Bosch Group, AI Research Intern — Bengaluru, India, Mar 2025 – July 2025, 5 months

- Researched weather-resilient ADAS perception for rain, fog, and low-visibility conditions; work resulted in three Indian patent applications.

The Bosch Group, Generative AI Intern — Bengaluru, India, June 2024 – Aug 2024, 3 months

- Built a generative-AI-powered chatbot for an upcoming BGSW Data Accelerator system, including metadata management and information retrieval.

The Bosch Group, Machine Learning and Computer Vision Intern — Bengaluru, India, June 2023 – July 2023, 2 months

- Developed computer-vision models for diesel-engine production-line automation and microscopic-part classification.

PES University - Department of Computer Science and Engineering, Teaching Assistant — Bengaluru, India, July 2024 – May 2025, 11 months

- Teaching Assistant for Generative AI and its Applications and Mathematics for Computer Science Engineers.
- Prepared slide decks and coding assignments, guided course projects, and evaluated assignments.

Publications and Patents

- **ICAART 2025:** "Enhancing Multi-Agent LLM Frameworks: A Context-Aware Approach for Securing Dynamic Agent Collaboration."
- **Patent Applications:** 202541105316, 202541093849, and 202641023935, from Bosch research on weather-resilient ADAS perception.

Technical Skills

Programming: Python, C, Java, SQL, MATLAB, R

ML / Research: PyTorch, JAX, Computer Vision, Representation Learning, World Models, JEPA, LLMs, Gaussian Processes

Tools: AWS, Docker, Kubernetes, Slurm